

3- Bayesian Inference (ii)

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Overview

- 1- Jeffreys priors
- 2- Bayesian hypothesis testing

1- Jeffreys prior

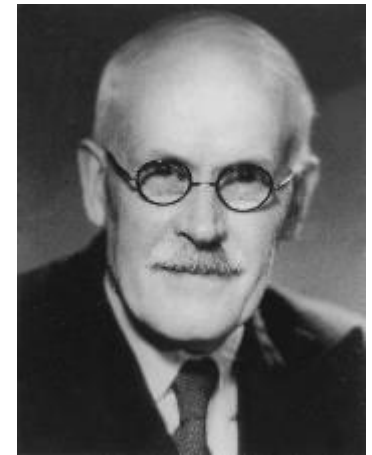
The Jeffreys prior is a non-informative (objective) prior distribution such that its density is proportional to the square root of the determinant of the Fisher information matrix:

$$\pi(\theta) \propto \sqrt{\det \mathcal{I}(\theta)}.$$

The Fisher information matrix is defined as, $\mathcal{I}(\theta) = E \left(\left(\frac{\partial}{\partial \theta} \log L(X|\theta) \right)^2 \right)$

If the likelihood function is twice differentiable with respect to θ , and under certain regularity conditions, then the Fisher information may also be written as,

$$\mathcal{I}(\theta) = -E \left(\frac{\partial^2}{\partial \theta^2} \log L(X|\theta) \right)$$



William H. Jeffreys

Exercise: Calculate the Jeffreys prior for the breast cancer screening study.

The likelihood function is, $L(X|\theta) = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}$

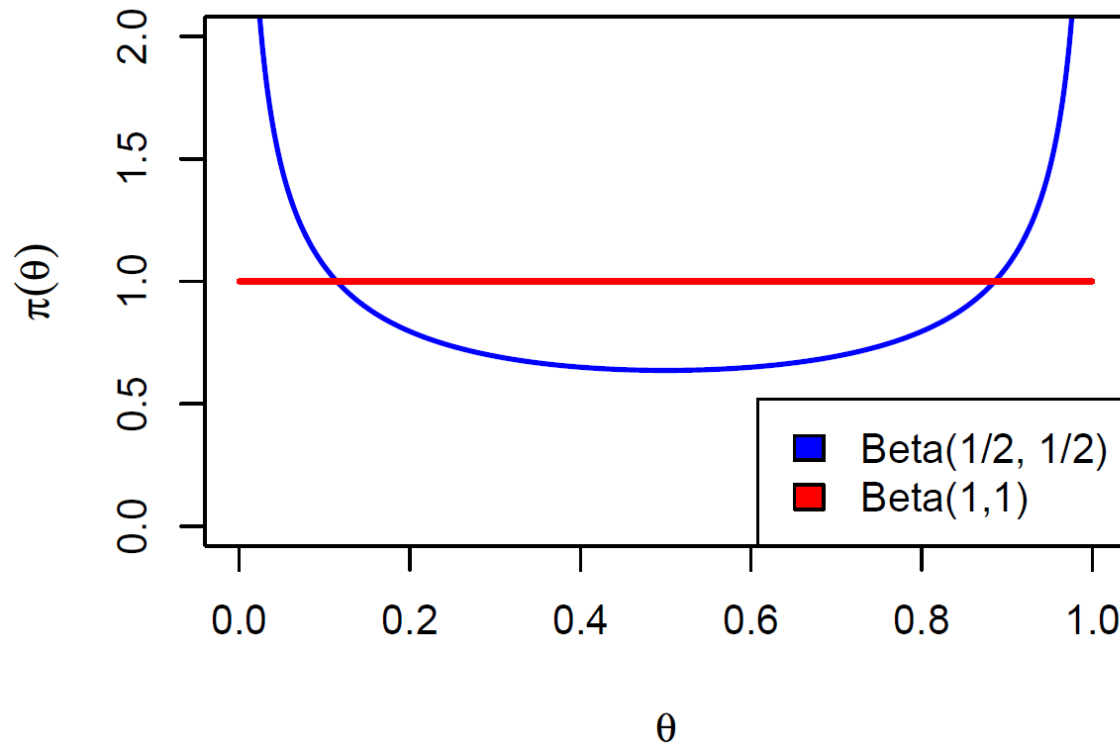
The log-likelihood function and its derivatives:

$$l(X|\theta) = \log(\theta)(\sum_{i=1}^n x_i) + \log(1 - \theta)(n - \sum_{i=1}^n x_i)$$

$$\frac{dl}{d\theta} = \frac{\sum_{i=1}^n x_i}{\theta} - \frac{n - \sum_{i=1}^n x_i}{1 - \theta} \quad \frac{d^2l}{d\theta^2} = -\frac{\sum_{i=1}^n x_i}{\theta^2} - \frac{n - \sum_{i=1}^n x_i}{(1 - \theta)^2}$$

$$I(\theta) = -E \left(\frac{d^2l}{d\theta^2} \right) = \frac{n\theta}{\theta^2} + \frac{n(1-\theta)}{(1-\theta)^2} = \frac{n}{\theta(1-\theta)}$$

$$\pi_J(\theta) \propto \theta^{-1/2}(1 - \theta)^{-1/2} \sim \text{Beta}(1/2, 1/2)$$



The Jeffreys prior favours values of θ for which $I(\theta)$ is large. The minimum information corresponds to $\theta=1/2$, that is, the maximum uncertainty.

1- An important property of the Jeffreys prior π_J .

For the breast cancer screening study, using the Jeffrey's prior, the posterior is:

$$f(\theta|X) \propto \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i} \theta^{-1/2} (1 - \theta)^{-1/2}$$

$$f(\theta|X) \sim \text{Beta}(\sum_{i=1}^n x_i + 1/2, 1/2 + n - \sum_{i=1}^n x_i)$$

Let us define a new parameter $\varphi = h(\theta)$ as a reparameterization (we are assuming that θ has dimension 1). For instance, we consider the change of parameter $\varphi = \theta/(1 - \theta)$ (this is the odds). What is the posterior density now?

The posterior on φ has the form, $f(\varphi) = f(g(\varphi))g'(\varphi)$, where $g(\cdot)$ is the inverse function of $h(\cdot)$ (check this!).

In this case, $g(\varphi) = \varphi/(1 + \varphi)$, so that,

$$f(\varphi|X) \propto \left(\frac{\varphi}{1+\varphi}\right)^{\sum_{i=1}^n x_i - 1/2} \left(1 - \frac{\varphi}{1+\varphi}\right)^{n - \sum_{i=1}^n x_i - 1/2} \frac{1}{(1+\varphi)^2}$$

$$f(\varphi|X) \propto \frac{\varphi^{\sum_{i=1}^n x_i - 1/2}}{(1+\varphi)^{n+1}}$$

For our particular example,
$$f(\varphi|X) = \frac{\varphi^{13.5}}{(1+\varphi)^{301}} \frac{\Gamma(301)}{\Gamma(14.5)\Gamma(286.5)}$$

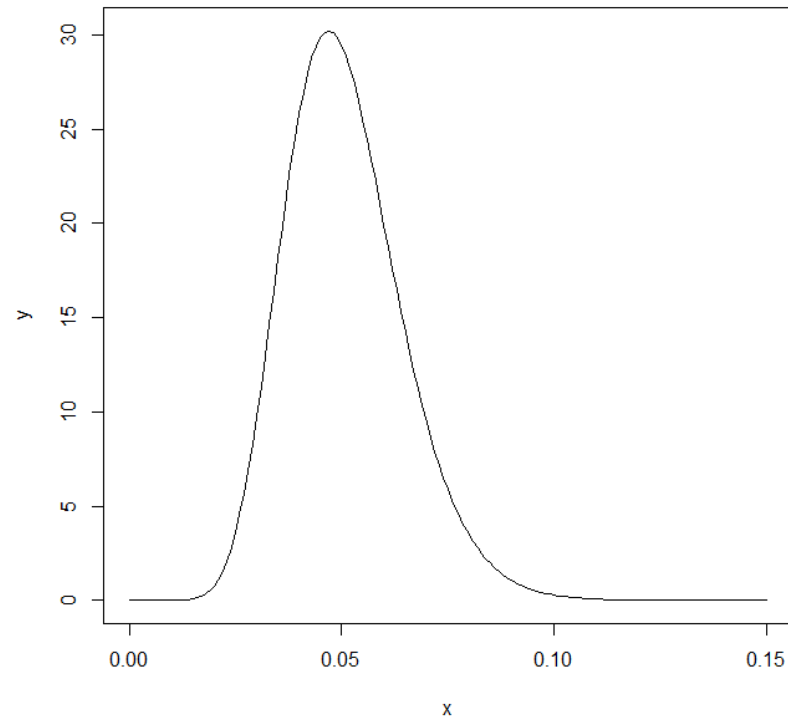
This is a density defined for $\varphi \geq 0$.

```
x=seq(0,0.15,0.001)
```

```
y=exp(13.5*log(x)-301*log(1+x)+lgamma(301)-lgamma(286.5)-lgamma(14.5))
```

```
plot(x,y,type="l")
```

By numerical integration we can see that the expectation is, $E(\varphi) = 0.0508$



2- An important property of the Jeffreys prior π_J .

We can also make the change of parameter from the beginning, that is,

The likelihood function is, $L(X|\varphi) = \left(\frac{\varphi}{1+\varphi}\right)^{\sum_{i=1}^n x_i} \left(1 - \frac{\varphi}{1+\varphi}\right)^{n - \sum_{i=1}^n x_i}$

$$\begin{aligned} l(X|\varphi) &= \log\left(\frac{\varphi}{1+\varphi}\right)\left(\sum_{i=1}^n x_i\right) + \log\left(1 - \frac{\varphi}{1+\varphi}\right)\left(n - \sum_{i=1}^n x_i\right) \\ &= \log(\varphi)\left(\sum_{i=1}^n x_i\right) - n \log(1 + \varphi) \end{aligned}$$

$$\frac{dl}{d\varphi} = \frac{\sum_{i=1}^n x_i}{\varphi} - \frac{n}{1+\varphi} \quad \frac{d^2l}{d\varphi^2} = -\frac{\sum_{i=1}^n x_i}{\varphi^2} + \frac{n}{(1+\varphi)^2}$$

$$I(\varphi) = -E\left(\frac{d^2l}{d\varphi^2}\right) = \frac{n\varphi}{(1+\varphi)\varphi^2} - \frac{n}{(1+\varphi)^2} = \frac{n}{\varphi(1+\varphi)^2}$$

$$\pi_J(\varphi) \propto \frac{1}{\varphi^{1/2}(1+\varphi)}$$

The posterior is,
$$f(\varphi|X) \propto \frac{\varphi^{\sum_{i=1}^n x_i - 1/2}}{(1+\varphi)^{n+1}}$$

We obtain the same result. It only happens for the Jeffreys prior !!

This is a kind of principle of invariance: one should be able to apply these priors to certain situations, apply a change of variable, and still get the same answer.

Exercises:

1-Show that for the Poisson distribution the Jeffreys prior is $\pi_J(\lambda) = 1/\lambda^{1/2}$

2-Calculate the Jeffreys prior for the Normal distribution with σ^2 known.

2- Bayesian hypothesis testing

In classical (frequentist) statistics the p-value is just the probability of rejecting the null hypothesis (H_0) given the null hypothesis is really true. It can be written in this way:

$$\text{p-value} = P(\text{Data} \mid H_0 \text{ is true})$$

However, the p-value is frequently misinterpreted as $P(H_0 \text{ is true} \mid \text{Data})$!!

But $P(H_0 \text{ is true} \mid \text{Data})$ is very interesting and meaningful. Bayesian inference allows to compute this probability and, in fact, this is the Bayesian procedure for hypothesis testing.

We want to test $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$

The distribution of the data given the parameter: $X|\theta \sim L(X|\theta)$

Prior distributions:

-Prior probabilities of the hypothesis, $P(H_0)$ and $P(H_1)$. An uninformative option is $P(H_0)=P(H_1)=0.5$.

-Prior densities $\pi_0(\theta)$ and $\pi_1(\theta)$ defined on Θ_0 and Θ_1 . If Θ_i is a point then the density $\pi_i(\theta)$ is a point mass.

Posterior probabilities: (an application of Bayes Theorem)

$$P(H_0|X) = \frac{P(H_0) \int_{\Theta_0} L(X|\theta) \pi_0(\theta) d\theta}{P(H_0) \int_{\Theta_0} L(X|\theta) \pi_0(\theta) d\theta + P(H_1) \int_{\Theta_1} L(X|\theta) \pi_1(\theta) d\theta} = 1 - P(H_1|X)$$

Example: Remember the breast cancer screening study, 14 of the 300 women had positive tests. We want to test,

$$H_0: \theta = \theta_0 \quad (\theta_0 = 0.036)$$

$$H_1: \theta \neq \theta_0$$

We assume that $P(H_0) = P(H_1) = 0.5$, and $\pi_1(\theta)$ to be uniform.

$$P(H_0|X) = \frac{\int_{\Theta_0} L(X|\theta) \pi_0(\theta) d\theta}{\int_{\Theta_0} L(X|\theta) \pi_0(\theta) d\theta + \int_{\Theta_1} L(X|\theta) \pi_1(\theta) d\theta} = \frac{L(X|\theta_0)}{L(X|\theta_0) + \int_{\theta \neq \theta_0} L(X|\theta) d\theta}$$

Note that due to continuity, $\int_{\theta \neq \theta_0} L(X|\theta) d\theta = \int_0^1 L(X|\theta) d\theta$

Then, $\int_0^1 L(X|\theta) d\theta = \int_0^1 \theta^s (1 - \theta)^{n-s} d\theta = \beta(s + 1, n - s + 1)$,
where $s = \sum_{i=1}^n x_i$

Consequently,

$\theta = 0.036; x = 14; n = 300$

$1/(1 + \theta^{-(x+1)}(1 - \theta)^{n-x}) \cdot \beta(x+1, n-x+1)$

$P(H_0|X) = 0.9541$



Exercise:

For the example of the breast cancer screening study,
Perform the following test,

$$H_0: \theta = 0.05$$

$$H_1: \theta < 0.05$$

-Calculate $P(H_0 | X)$ and $P(H_1 | X)$.

The Bayes factor

We are interested in testing H_0 against H_1 . In order to make a decision we calculate the proportion,

$$\frac{Pr(H_0|X)}{Pr(H_1|X)} = \frac{Pr(H_0|X)}{1 - Pr(H_0|X)}$$

This proportion is known as *Bayes factor*. This is just the posterior odds in favor of H_0 over H_1 . If we have that $P(H_0)=P(H_1)=0.5$ and the hypothesis are simple this is just a likelihood ratio.

Note that for the breast cancer screening study, $P(H_0|X) = 0.9541$ and the Bayes factor is, 20.786. What is the conclusion?

Table 1. Jeffreys' scale of evidence.

$B_{12} \in$	Evidence against M_2
(1,3.2)	Not worth more than a bare mention
(3.2,10)	Substantial
(10,100)	Strong
(100, ∞)	Decisive.

Table 2. Scale of evidence proposed by Kass and Raftery (1995).

$B_{12} \in$	Evidence against M_2
(1,3)	Not worth more than a bare mention
(3,20)	Positive
(20,150)	Strong
(150, ∞)	Very strong.

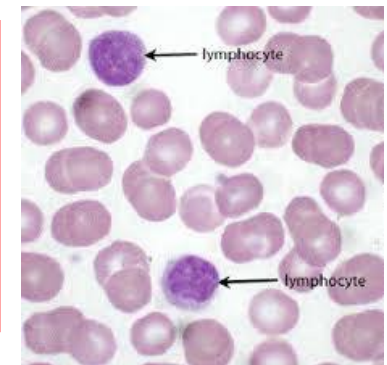
In these tables M_2 is our H_1 . Although these rules are given in terms of evidence against H_1 , the same rule can be used to interpret the evidence against H_0 by inverting the value of the Bayes factor. Here the Bayes factor is indicated as B_{12} .

Example 1

A clinical test is performed by means of a blood analysis, where the number of lymphocytes are counted in a certain number of squares. An individual is considered healthy if the mean is $H_0: 2 \leq \mu \leq 4$ and infected if $H_1: \mu > 4$. The result of a clinical test has given a sample mean of 4.6 with a sample of 100 squares and a total lymphocyte count of 459. Assuming that the counts are Poisson distributed, what is the conclusion?

Remember that for Poisson observations the likelihood function is,

$$\begin{aligned} L(\lambda | x_1, \dots, x_n) &\propto P(x_1, \dots, x_n | \lambda) \\ &\propto \lambda^{\sum_{i=1}^n x_i} e^{-n\lambda} \end{aligned}$$



Assuming that $P(H_0)=P(H_1)=0.5$, the Bayes factor is,

$$\frac{Pr(H_0|X)}{Pr(H_1|X)} = \frac{\int_2^4 L(\mu|X)\pi_0(\mu) d\mu}{\int_4^\infty L(\mu|X)\pi_1(\mu) d\mu}$$

Assuming $\pi_0(\mu)=\pi_1(\mu) \sim \text{Gamma}(\alpha,1)$, where the hyper-parameter α is estimated by the sample mean (4.6), the Bayes factor is,

$$\frac{Pr(H_0|X)}{Pr(H_1|X)} = \frac{\int_2^4 \mu^{463.6} \exp(-101\mu) d\mu}{\int_4^\infty \mu^{463.6} \exp(-101\mu) d\mu}$$

Numerical calculations can be problematic. However, dividing numerator and denominator by an appropriate constant this is just, $(F(4)-F(2))/(1-F(4))$ where F is the distribution function of a $\text{Gamma}(464.6, 101)$.

It gives a Bayes factor of,

> (pgamma(4,464.6,101)-pgamma(2,464.6,101))/(1-pgamma(4,464.6,101))
[1] 0.001707005

It gives an evidence against H_0 of $1/0.001707 = 585.8231$.

What is the required sample size in order to obtain an evidence against $H_0 \geq 100$, when the sample mean is ≥ 5 ?

Now the Bayes factor, as a function of the sample size n , is the following:

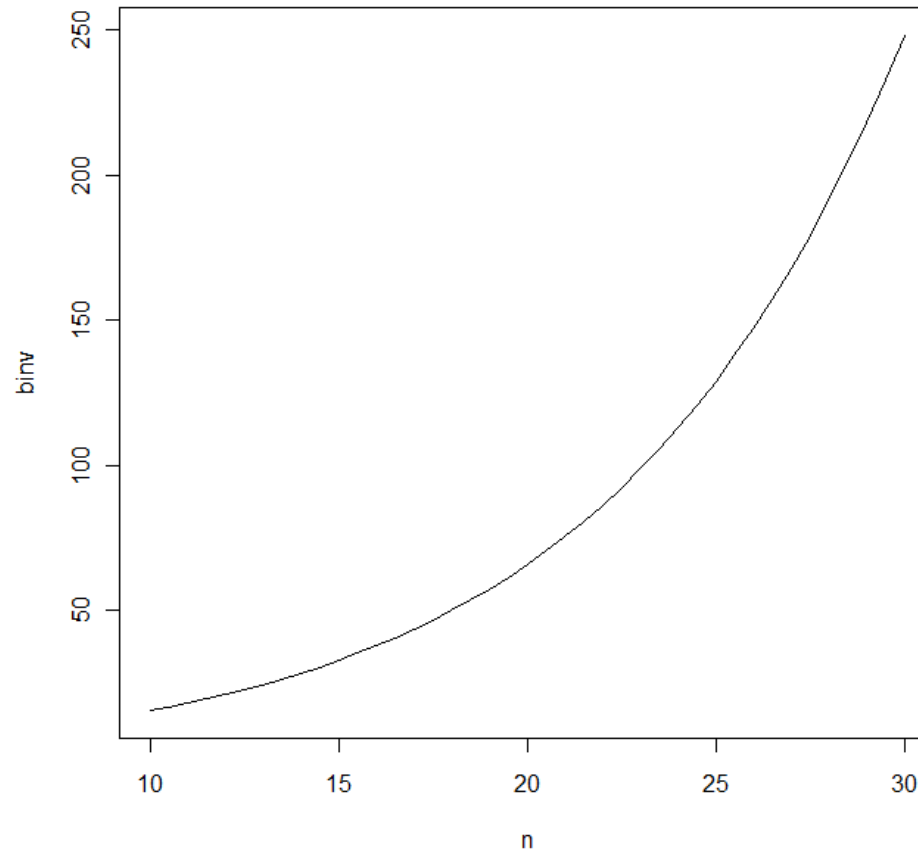
$$\frac{Pr(H_0|X)}{Pr(H_1|X)} = \frac{\int_2^4 \mu^{5n+4} \exp(-n-1) d\mu}{\int_4^\infty \mu^{5n+4} \exp(-n-1) d\mu}$$

Again, dividing numerator and denominator by an appropriate constant this is just, $(F(4)-F(2))/(1-F(4))$ where F is the distribution function of a $\text{Gamma}(5(n+1), n+1)$.

We plot the inverse of the Bayes factor against n in order to explore what happens:

```
n=seq(10,30,0.5)
binv=(1-pgamma(4,5*(n+1),n+1))/(pgamma(4,5*(n+1),n+1)-pgamma(2,
5*(n+1),n+1))
plot(n,binv,type="l")
```

The conclusion is that a sample size about 25 can be enough.



Example 2

A subject is asked to guess the colour (red or black) of each of 20 randomly selected playing cards. The subject correctly guesses 15 colours. Does this level of success indicate that she has extra-sensory perception (ESP)?

Under the hypothesis H_0 that he/she does not have ESP, the probability of correctly identifying 15 out of 20 colours is $L(X | H_0) = 0.0148$ (check this!).



If he/she does have ESP (the other hypothesis H_1), then we suppose that she has a probability $\theta > 0.5$ of correctly guessing each card. Under the assumption that all values of θ from 0.5 to 1 are equally likely (uniform prior distribution), the probability that the medium correctly identifies 15, given that he/she does have ESP, is

```
integrand= function(t) {dbinom(15,20,t)*2}  
integrate(integrand,0.5, 1)$value
```



0.0940

The Bayes factor in favor of H_1 is $0.0940/0.0148 = 6.351$.

This is clearly not strong evidence against H_0 , and would not be enough to persuade most people to believe in ESP.

(Perform a classical test and compare the results.)



What happens if the prior distribution is changed?

Exercise: Prove that, based on the data of 15 correct answers out of 20, then the posterior odds against H_0 will always be lower or equal than 13.7.

Exercise:

The on-call time in a telephone-based service follows an exponential distribution with rate θ (the rate is the reciprocal of the expectation). We have observed eight users of this service and their on-call time average has been 1.5 minutes.

- Test $H_0: \theta=0.6$ and $H_1: \theta=0.8$, being non-informative. Compute the Bayes Factor and explain the results.
- Test $H_0: \theta=0.6$ against $H_1: \theta \neq 0.6$, being non-informative and explain the results (chose for H_1 the improper prior $\pi(\theta) = 1$).